Limits Graphically and Numerically

ANSWER KEY

Estimating limits from a table

1 A function g satisfies $g(2.9) = 6.7$, $g(2.99) = 6.97$, $g(2.999) = 6.997$, $g(3.1) = 7.3$, $g(3.01) = 7.03$, $g(3.001) = 7.003$. Estimate $\lim_{x \to 3} g(x)$.

Both sides approach 7: $\lim_{x \to 3} g(x) = 7$.

2 Using a table of values with x near 0 (radians), estimate $\lim_{x \to 0} \frac{1 - \cos x}{x^2}$ to two decimal places (the known exact value is $\frac{1}{2}$).

Values approach 0.50 from both sides, consistent with the exact limit $\frac{1}{2}$.

Reading limits from a graph

3 The graph below shows $f(x) = \frac{x^2 - 1}{x - 1}$, which is undefined at $x = 1$. Based on the simplified form $f(x) = x + 1$ $(x \neq 1)$, state $\lim_{x \to 1} f(x)$.

$\lim_{x \to 1} f(x) = 2$ (even though $f(1)$ itself is undefined — a removable discontinuity).

4 A function h has $\lim_{x \to 2^-} h(x) = 5$ and $\lim_{x \to 2^+} h(x) = 5$, but $h(2) = 8$. State $\lim_{x \to 2} h(x)$ and explain whether h is continuous at $x = 2$.

$\lim_{x \to 2} h(x) = 5$, since both one-sided limits agree. But h is NOT continuous at $x = 2$, since $\lim_{x \to 2} h(x) = 5 \neq h(2) = 8$ — this is a removable discontinuity.

One-sided limits and limits that fail to exist

5 For $f(x) = \frac{x}{|x|}$, find:

a $\lim_{x \to 0^+} f(x) = 1$ **b** $\lim_{x \to 0^-} f(x) = -1$ **c** $\lim_{x \to 0} f(x)$ does not exist

6 A piecewise function is defined by $f(x) = x^2$ for $x < 1$, and $f(x) = 3 - x$ for $x \geq 1$. Find $\lim_{x \to 1^-} f(x)$, $\lim_{x \to 1^+} f(x)$, and state whether $\lim_{x \to 1} f(x)$ exists.

$\lim_{x \to 1^-} f(x) = 1^2 = 1$. $\lim_{x \to 1^+} f(x) = 3 - 1 = 2$. Since the one-sided limits disagree $(1 \neq 2)$, $\lim_{x \to 1} f(x)$ does not exist.

Algebraic evaluation

7 Evaluate each limit by direct substitution or factoring:

a $\lim\limits_{x\to 4}(3x^2 - 5x + 1) \quad = 3(16) - 20 + 1 = 29$

b $\lim\limits_{x\to 3}\frac{x^2 - 9}{x - 3} \quad = \lim\limits_{x\to 3}(x+3)=6$

c $\lim\limits_{x\to -2}\frac{x^2 + 5x + 6}{x + 2} \quad = \lim\limits_{x\to -2}(x+3)=1$

8 Evaluate $\lim\limits_{x\to 0}\frac{\sqrt{x+9}-3}{x}$ by rationalizing the numerator.
Multiply by $\frac{\sqrt{x+9}+3}{\sqrt{x+9}+3}$: $\frac{x}{x(\sqrt{x+9}+3)} = \frac{1}{\sqrt{x+9}+3} \to \frac{1}{6}$.

Limits at infinity from a graph

9 The graph shows $f(x) = \frac{2x}{x+1}$, which has a horizontal asymptote. Use the graph to estimate $\lim\limits_{x\to\infty}f(x)$, then confirm algebraically by dividing numerator and denominator by x.
The graph flattens toward height 2 as $x \to \infty$. Algebraically: $\frac{2x}{x+1} = \frac{2}{1+1/x} \to \frac{2}{1+0} = 2$.

Limits involving infinite discontinuities

10 For $f(x) = \frac{1}{x-3}$, find $\lim\limits_{x\to 3^-}f(x)$ and $\lim\limits_{x\to 3^+}f(x)$, and state whether $\lim\limits_{x\to 3}f(x)$ exists.
As $x \to 3^-$, $x - 3 \to 0^-$, so $f(x) \to -\infty$. As $x \to 3^+$, $x - 3 \to 0^+$, so $f(x) \to +\infty$. Since the one-sided limits are not equal (in fact neither is finite), $\lim\limits_{x\to 3}f(x)$ does not exist.

11 Evaluate $\lim\limits_{x\to\infty}\frac{3x^2 + 1}{x^2 - 4}$, using the fact that the numerator and denominator have the same degree.
For equal-degree rational functions, the limit is the ratio of leading coefficients: $\lim\limits_{x\to\infty}\frac{3x^2 + 1}{x^2 - 4} = \frac{3}{1} = 3$.

Estimating limits with a calculator-style table

12 Construct a mental table of values for $f(x) = \frac{e^x - 1}{x}$ at $x = -0.1, -0.01, 0.01, 0.1$ (all close to 1), and use it to estimate $\lim\limits_{x\to 0}\frac{e^x - 1}{x}$.
The values all cluster near 1 (e.g. $f(0.1) \approx 1.052$, $f(-0.1) \approx 0.952$, both approaching 1 as $x \to 0$): $\lim\limits_{x\to 0}\frac{e^x - 1}{x} = 1$ (a standard limit related to the derivative of e^x at 0).

Limits with piecewise graphical behavior

13 A graph shows $f(x) = 2$ for $x < 0$, $f(x) = x + 2$ for $0 \le x < 3$, and $f(x) = 8 - x$ for $x \ge 3$. Find $\lim\limits_{x\to 0}f(x)$ and $\lim\limits_{x\to 3}f(x)$, stating whether each exists.
At $x = 0$: $\lim\limits_{x\to 0^-}f(x) = 2$, $\lim\limits_{x\to 0^+}f(x) = 0 + 2 = 2$: limit exists, equals 2. At $x = 3$: $\lim\limits_{x\to 3^-}f(x) = 3 + 2 = 5$, $\lim\limits_{x\to 3^+}f(x) = 8 - 3 = 5$: limit exists, equals 5.

14 Evaluate $\lim_{x \to 2} \frac{x^2 - 5x + 6}{x^2 - 4}$ by factoring both the numerator and denominator.

$\frac{(x-2)(x-3)}{(x-2)(x+2)} = \frac{x-3}{x+2} \to \frac{2-3}{2+2} = -\frac{1}{4}$.